Food Safety Risk Patterns in Chicago

Inspection Outcomes by Facility and ZIP Context

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Main question

Which factors are associated with higher food inspection failure risk in Chicago?

- Focus on overall inspection results rather than ranking individual restaurants.
- Compare failure patterns by facility type, risk level, and ZIP-code context.
- Help public health stakeholders identify where inspection failure risk appears higher.
- Give everyday consumers a clearer picture of general food safety risk in Chicago.

Data

Chicago Food Inspections

- Data size: 129,274 inspection records from 2019–2026
- Outcome: inspection result (Pass, Pass with Conditions, Fail)
- Features: facility type, risk level, ZIP code, etc.

Census ACS ZIP Code Context

- ZIP-code level socioeconomic data from the Census ACS API
- Includes poverty rate, median income, education, population, and age composition
- Joined two datasets by ZIP code

EDA Finding 1: Outcomes and Risk Levels

Overall outcomes

- Pass: 44.90%
- Pass w/ conditions: 19.73%
- Fail: 19.21%

Risk level

Risk level is based on food handling type, not past inspection outcome. e.g.

- Risk 1: restaurant
- Risk 2: coffee shop
- Risk 3: convenience store

Failure by risk

- Risk 1: 19.38%
- Risk 2: 18.45%
- Risk 3: 19.17%

Interpretation

The risk label is useful context, but it does not strongly separate pass/fail outcomes.

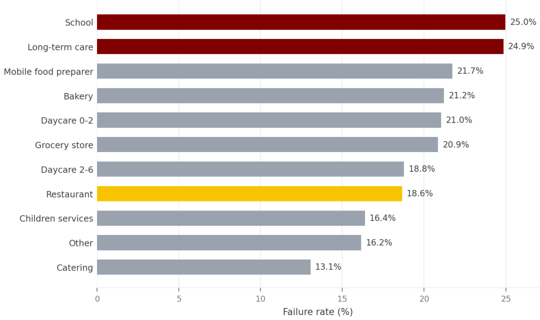
EDA Finding 2: Failure Rates by Facility Type

Failure rates differ by facility category.

Key finding

Schools and long-term care facilities have the highest observed failure rates, both near 25%.

Restaurants account for most inspections, but their failure rate is lower, around 18.6%.



Takeaway

Food safety risk is not only about restaurants; schools and long-term care facilities also deserve attention because they serve more vulnerable groups.

EDA Finding 3: ZIP Code Context

Inspection failure rates vary with ZIP-code socioeconomic context.

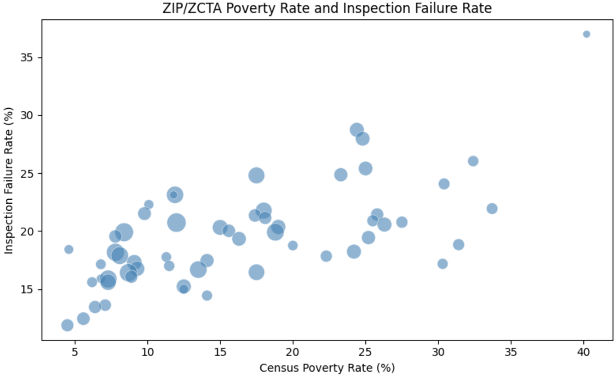
Key pattern

ZIP-level poverty, income, and education are all associated with observed inspection failure rates at ZIP code level.

Correlations with failure rate:

- Poverty: $r = 0.665$
- Income: $r = -0.633$
- Bachelor's: $r = -0.607$

Association only; not a causal neighborhood claim.



Modeling Results

I use a logistic regression model to predict inspection failure risk.

Model question

Can facility type, risk level, inspection type, time, and ZIP-code context help identify inspections with higher failure risk?

The table compares inspections with the lowest and highest predicted probabilities of failure.

Model-ranked group	Actual failure rate
Lowest predicted risk	7.9%
Highest predicted risk	29.6%

Interpretation

The highest-risk group fails almost four times as often as the lowest-risk group. This gap suggests that the model can help identify groups of inspections with higher failure risk.

Conclusion

Food inspection failure risk is shaped by multiple layers of context.

- Overall, about one in five inspections fail, but Risk 1, Risk 2, and Risk 3 have very similar failure rates.
- Facility type shows clearer differences: schools and long-term care facilities have the highest observed failure rates.
- ZIP-code socioeconomic context is also associated with failure rates, especially poverty, income, and education.
- The logistic model brings these factors together and separates inspections into groups with different actual failure rates.